

JMO – 2001

Max Mark – 100

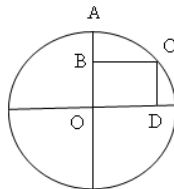
Time – 3 Hrs

Instructions:

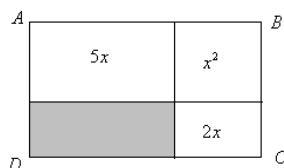
- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all questions.
- Group A each question contains $2\frac{1}{2}$ marks and Group B each question contain 5 marks each.

Group - A

1. Which number is half way between 179 and 837?
2. $2\sqrt{2}$ is what percentage of $8\sqrt{8}$?
3. I have two brothers and two sisters. My mother's parents have 12 grand children and my father's parents have 15 grand children. What is the total number of grand children my maternal nad paternal grandparents have?
4. What is the largest natural number less than 2001 that is divisible by 3 and the sum of whose is divisible by 5?
5. The product of n consecutive integers starting from 1 is denoted by n!. For example: $12! = 1 \times 2 \times \dots \times 12$. Now if $n! = 3! \times 5! \times 7!$. Find the value of n.
6. If the number of books in the library of Institute of Mathematics and Application is a number between 1961 and 2001. Exactly $\frac{1}{7}$ th of them are novels and exactly 20% of them are poetry books. How many books are there in the library?
7. If $x^p = \sqrt[3]{x \times \sqrt[4]{x}}$, then write the value of p.
8. Karina received 60% of votes in an election in which Hritik and Amitav contested with her. The ration of Hritik polled to that of Amitav was 5:7. If Amitav received 28 more votes than Hritik how many votes did Karina received?
9. If $m * n = m^3 - 2n$ then find $3 * 2 = ?$
10. Add the following $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots$ upto 20 terms.
11. Sandeep earns 20% nore than Hema. By how much percent does Hema eans less than Sandeep?
12. Express the following into lowest form: $\frac{97619}{129673}$.
13. See the figure O is the centre of the circle. OBCD is a rectangle with OB = 19, BC = 90, find AB = ?



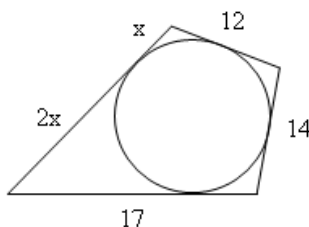
14. D is an integer greater than 1. Math teacher Namita divided 1996 by D to get remainder R. Mathe teacher Sabita divided 1789 by D to get the same remainder R. Math teacher Indulata divided 1053 by D to get again the same remainder R. Find out the largest possible value for D and also find out the value of R.
15. ABCD is a rectangle. The areas of 3 smaller enclosed rectangles are given in the figure. Find out the area of shaded portion.



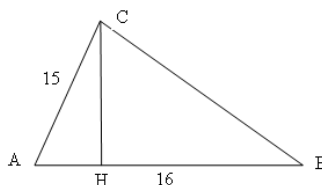
16. Find the number of diagonals of an octagon.

Group - B

17. In the triangle ABC, $AB = 2$, $BC = 5$, $AC = 4$. Find the length of height drawn from A to BC.
18. If A sells an article to B at a profit of 10% and B sells it to C at a profit of 10%, then at what % loss or profit must C sell the article to A so that A gets its back at the original price at which it was purchased by him?
19. What is the sum of numbers less than 100 which are divisible by 6 but not 4?
20. A quadrilateral is circumscribed about a circle as shown in the figure. Find the value of x .



21. A right triangle ABC with hypotenuse AB has side $AC = 15$. Altitude CH divides AB into segment AH and BH with $HB = 16$. Find the area of triangle ABC.



22. Three packets are labeled. The packet labeled as W contains two pieces of white chalks, the packet labeled as R contains two pieces of red chalks where as WR contains one piece of white chalk and one piece of red chalks. The labels are interchanged such that the chalk inside the packets are different from what label on it suggest. You are allowed to take out one piece of chalk from any of these packets and say which packet contains what type of chalks. Describe your inference.
23. Two cyclists 50 km apart if they start at same time after 4 hours when they travel in the same direction and after 2 hours when they travel in opposite direction. Find their speed per hour.
24. By what number should the number 999 be multiplied so that the product number will consist of only ones?
25. Arrange the following in descending order: $x = \frac{111110}{111111}$, $y = \frac{222221}{222223}$, $z = \frac{333331}{333334}$.
26. A two digit number is divided by the sum of its digits. What is the largest possible remainder?
27. A printer printed number of pages of a book starting from 1 and uses 3198 digits in all. How many pages does the book have?
28. Person A has 16 sapphires, person B has 10 rubies, and person C has 8 diamonds. Each of them gives the other person two of his original stones. Now they each have a collection of stones of Rs. 270000. What is the value of one diamond?

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